

Exercise 1

Given is the following task set.

process i	phase Φ_i	computation time C_i	period T_i	deadline D_i
τ_1	0	2	5	5
τ_2	0	4	7	7

→ RT Analysis?

- a) Can schedulability be guaranteed under RM or EDF?
b) Try to find a schedule with EDF and RM.

Theorem:

A set of periodic tasks $\tau_1, \dots, \tau_n$ (with $D_i = T_i$) is schedulable with EDF iff $\sum_{i=1}^n C_i / T_i \leq 1$.

Any set of n periodic tasks with a processor utilization factor $\leq U_{lub} = n(2^{1/n} - 1)$ can be scheduled by RM.

Any set of periodic tasks with a processor utilization factor $\leq \ln 2$ can be scheduled by RM.

This is a sufficient condition, but not a necessary condition!

$$a) u = \frac{2}{5} + \frac{4}{7} = \frac{14 + 20}{35} \quad \text{EDF} \checkmark$$

$$RM^x = \frac{34}{35} \leq 1$$

$$\neq 2 \cdot (\sqrt[3]{2} - 1)$$

$$\approx 0.83$$

$\frac{1}{5} \quad \frac{1}{7}$
 $\text{prio}(\tau_1) > \text{prio}(\tau_2)$ (smaller period first)

Because not necessarily unsched.
with RM $\rightarrow$ Response Time An.

$$R_i^{(0)} = C_i \leq D_i$$

$$R_i^{(j+1)} = C_i + \sum_{k=1}^{i-1} \left\lceil \frac{R_i^{(j)}}{T_k} \right\rceil \cdot C_k \leq D_i$$

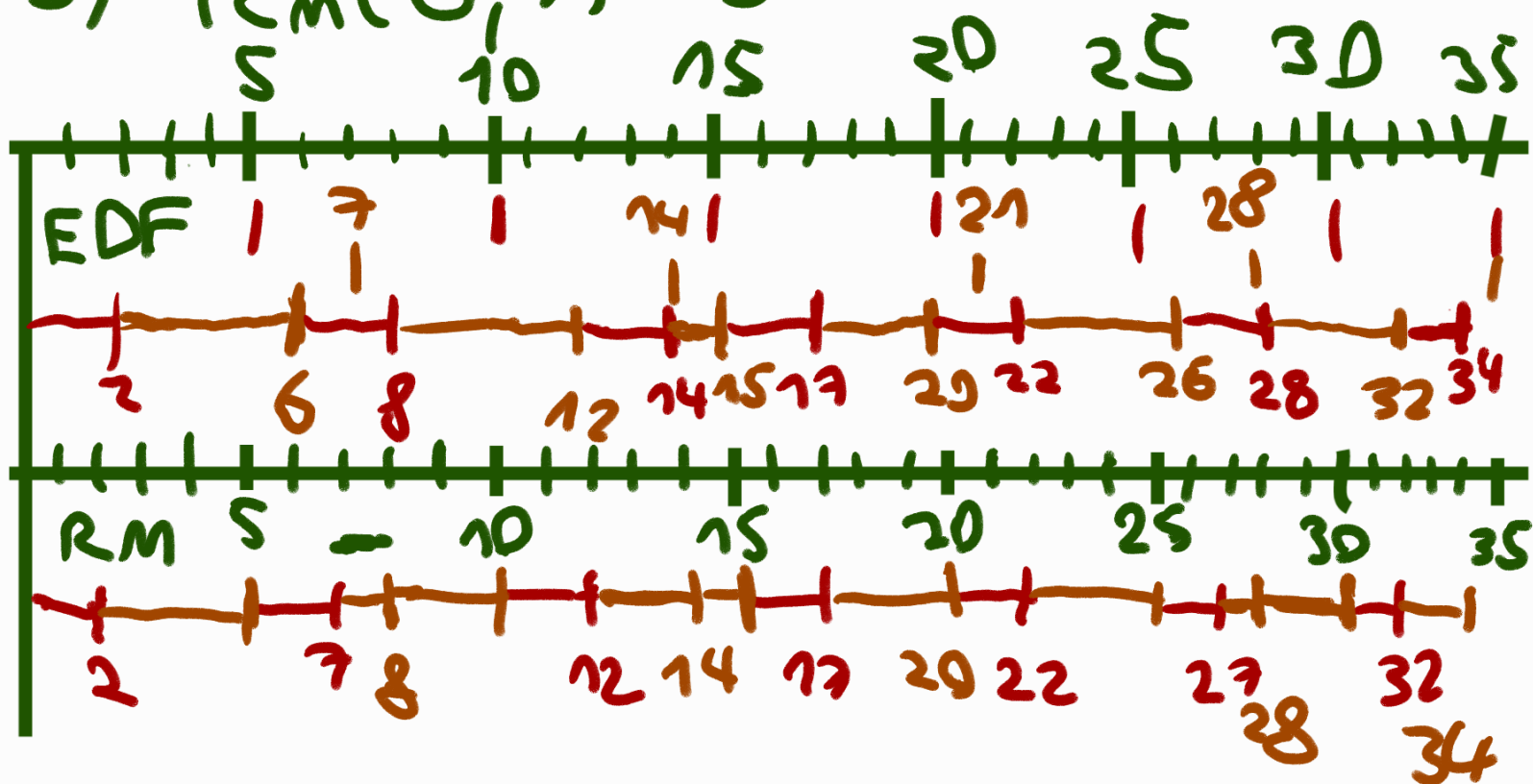
$$R_1^{(0)} = 2 \leq 5 = D_1$$

$$R_2^{(0)} = 4 \leq 7 = D_2$$

$$\begin{aligned} R_2^{(1)} &= 4 + \left\lceil \frac{4}{5} \right\rceil \cdot 2 = 6 \leq 7 \\ &= 4 + \left\lceil \frac{6}{5} \right\rceil \cdot 2 = 8 \not\leq 7 \end{aligned}$$

$\downarrow$
not
schedulable

$$b) \text{lc}_m(5, 7) = 35$$



Exercise 3

Given is the following task set of periodic tasks

process i	phase Φ_i	computation time C_i	period T_i	deadline D_i
τ_1	0	3	10	10
τ_2	0	2	4	4

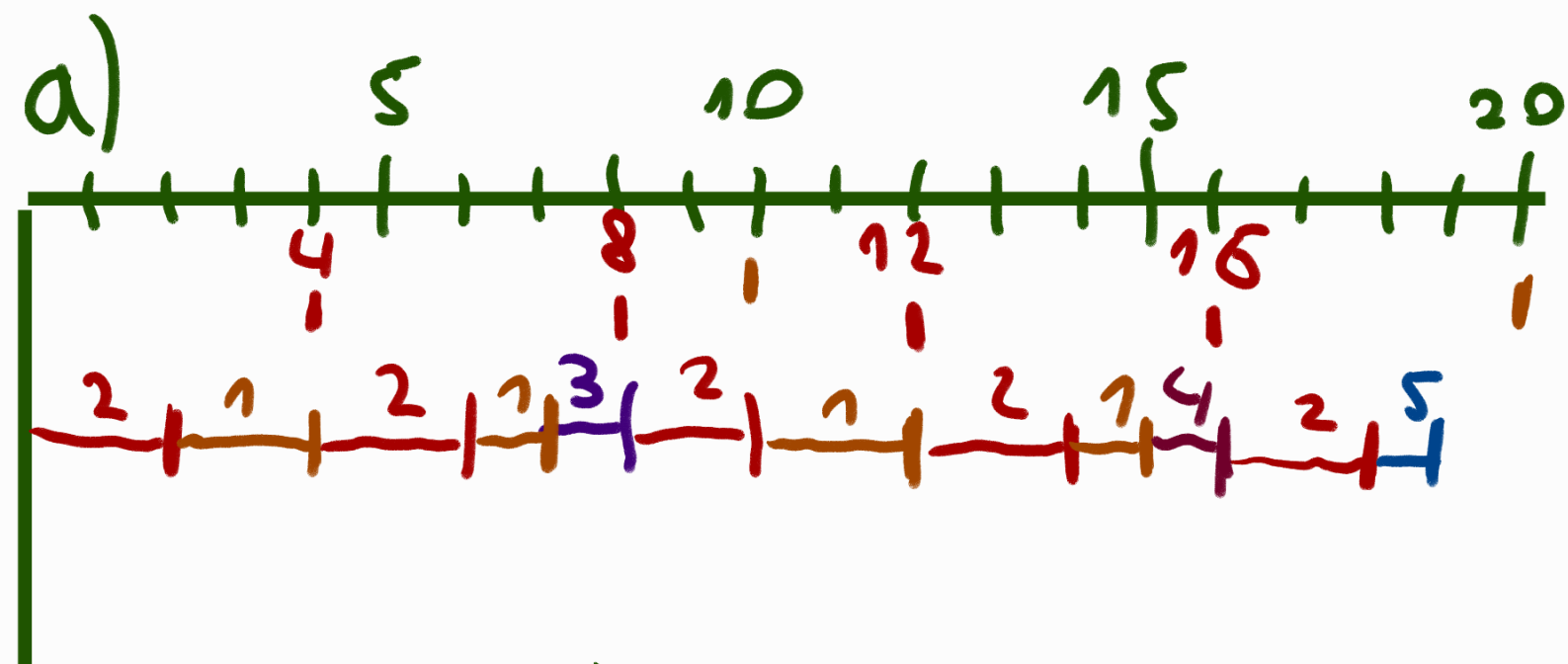
Also given is the following task set of aperiodic tasks

process i	arrival time a_i	computation time C_i
τ_I	2	1
τ_{II}	5	1
τ_{III}	9	1

- Give a schedule when the periodic tasks are scheduled using EDF and the aperiodic tasks are scheduled in the background.
- Compute the values of U_p and U_s according to the lecture.
- Compute the deadlines of the aperiodic tasks using the Total Bandwidth Server (TBS) algorithm.
- Schedule both the periodic and aperiodic tasks with deadlines computed according to TBS together using the EDF algorithm.

own priority: $\text{prio}(\tau_I) > \text{prio}(\tau_{II}) > \text{prio}(\tau_{III})$
 doesn't work for RM?
 ?

$$\text{lc}_m(10, 4) = 20$$



b)

$$U_p = \frac{3}{10} + \frac{2}{4} = \frac{6+10}{20} = \frac{16}{20} = \frac{4}{5}$$

$$U_s = 1 - \frac{4}{5} = \frac{1}{5}$$

c)

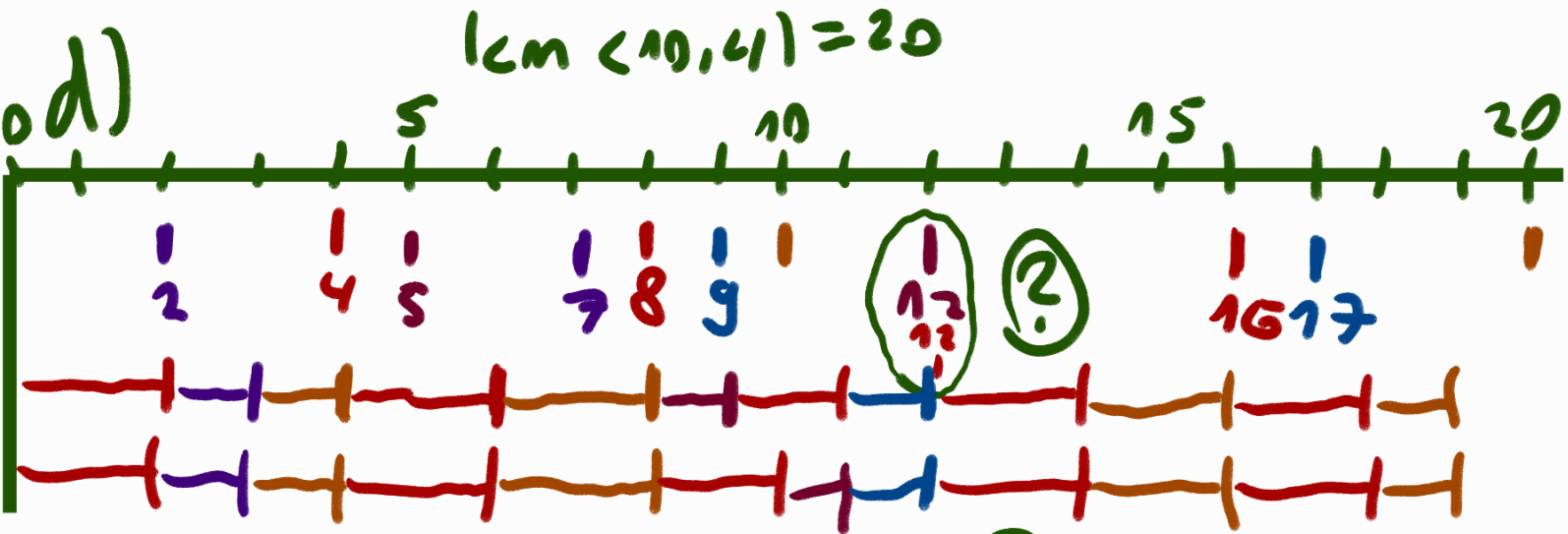
↳ sparse

$$d_{r_I} = 2 + \frac{1}{\frac{1}{5}} = 2 + 5 = 7$$

$$d_k = \max(r_k, d_{k-1}) + \frac{c_k}{U_s}$$

$$d_{r_{II}} = \max(7, 5) + 5 = 12$$

$$d_{r_{III}} = \max(12, 9) + 5 = 17$$



Exercise 4

We have to design a system that schedules periodic tasks with EDF and employs a total bandwidth server to serve aperiodic requests. We know of that the first aperiodic request has a computation time $C_a = 2$ and a relative deadline $D_a = 7$. What is the maximum processor utilization available for periodic tasks if we want to use the results about TBS to guarantee that this aperiodic task completes within its deadline.

$\Rightarrow RM(?)$

$$d_a = c_a + D_a$$

$$d_a + D_a = c_a + \frac{c_a}{U_s}$$

$$\Rightarrow 9 + 7 = 9 + \frac{2}{U_s}$$

$$\Rightarrow U_s = \frac{2}{7}$$

$$U_p \text{ maximal if } U_p = 1 - U_s = \frac{5}{7}$$

Exercise 2

The feasibility of a set of tasks with relative deadlines less than or equal to periods can be guaranteed with EDF by verifying that

$$\forall L > 0 \quad \sum_{i=1}^n \left\lfloor \frac{L + T_i - D_i}{T_i} \right\rfloor C_i \leq L.$$

Prove this theorem *only* for relative deadlines equal to periods. Note, use the theorem of the lecture: A set of periodic tasks $\tau_1, \dots, \tau_n$ (with $D_i = T_i$) is schedulable with EDF iff $\sum_{i=1}^n \frac{C_i}{T_i} \leq 1$. (1)

$$(1): \sum_{i=1}^n \frac{C_i}{T_i} \leq 1$$

$$(2): \text{if } \forall L > 0, \sum_{i=1}^n \left\lfloor \frac{L + T_i - D_i}{T_i} \right\rfloor C_i \leq L \quad \text{with } D_i = T_i$$

$$\Rightarrow: (1) \Rightarrow (2)$$

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$$\text{if } U \leq 1$$

$$\forall L > 0: L \geq UL = \sum_{i=1}^n \left(\frac{C_i}{T_i} \right) \cdot L$$

↑
Trans.

$$= \sum_{i=1}^n \left(\frac{L}{T_i} \right) \cdot C_i$$

$$\hookrightarrow \geq \sum_{i=1}^n \left(\frac{L}{T_i} \right) \cdot c_i$$

$$\Leftarrow: (2) \Rightarrow (1) \equiv \neg C_1 \rightarrow \neg C_2$$

If $U > 1$, then

$$\begin{aligned} \exists x L > 0 : L < UL &= \sum_{i=1}^n \left(\frac{c_i}{T_i} \right) \cdot L \\ &= \sum_{i=1}^n \left(\frac{L}{T_i} \right) \cdot c_i \\ &= \sum_{i=1}^n \left(\frac{L}{T_i} \right) \cdot c_i \end{aligned}$$

for $L = \text{lcm}(T_1, \dots, T_n)$